Type 1: Research Project

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MOF Graphs and Crystal Morphology: Crystal Shapes of MOFs from the Unit Cell

23CSE498 – Project Phase II (External Review) | **Team Number: G070**

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Guide Review & Submission Approval

Confirmation of Project Review and Approval from Guide

[Insert Screenshot of Guide Approval Email Here]

To be added by the team after the guide's approval.

1. Problem Definition & Research Motivation

Problem Statement, Identified Literature Gap & Objectives

Problem

- **Panel's question (Review 1):** "Is it possible to generate at least a few morphologies from the given unit cell structure?"
- **Why it matters:** a gas enters a crystal through its surface; the shape decides which faces are exposed and how much surface there is per volume.

Literature and gap

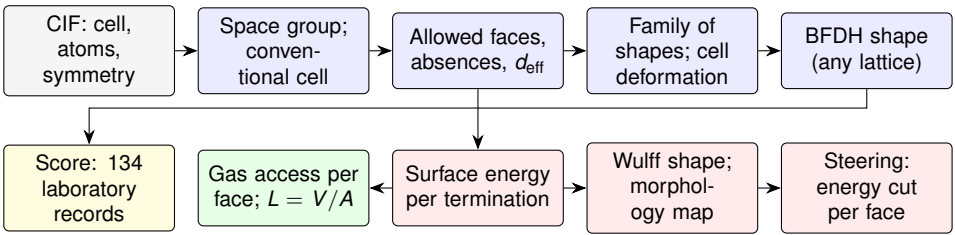
- BFDH (geometry), Wulff (energy), Hartman–Perdok (bond chains), CHGNet (learned potential) exist separately.
- **Gap:** no checked route from a MOF's CIF to its allowed faces, its family of shapes, a structure-only prediction scored against many laboratory crystals, and the effect on gas access.

Objectives

1. Allowed faces and shape families from the space group, with proofs.
2. A favoured shape from geometry and energy, scored against 134 laboratory records with baselines.
3. Steering and gas-access indicators per shape, with stated limits.

2. Proposed Methodology & Overall Workflow

From a CIF to a scored crystal shape



Two independent implementations

Python (`code/morphology/`) and a browser engine (`website/`), written from the same specification and compared number by number in 15 test suites.

3. Methodological Components & Formulation

Governing equations, dataset and metrics

Formulation

- Absent (hkl): $\mathbf{h}R = \mathbf{h}$ and $\mathbf{h} \cdot \mathbf{t} \notin \mathbb{Z}$; $d_{\text{eff}} = d_{hkl}/n$.
- BFDH: face distance $\propto 1/d_{\text{eff}}$.
- $\gamma(s) = \frac{1}{2}(d/V) \sum \varepsilon_{\text{pair}}\, c_{\text{pair}}(s)$; each cut bond 70 kcal/mol; van der Waals 2–7%.
- Cubic map: $x = \gamma_{110}/\gamma_{100}$, $y = \gamma_{111}/\gamma_{100}$; cube $y \geq \sqrt{3}$, $x \geq \sqrt{2}$; octahedron $y \leq 1/\sqrt{3}$.
- Access: probe 1.65 Å (CO_2), 1.82 Å (N_2); $L = V/A$.

Dataset

- 134 records, 35 papers, 12 MOFs: shape, faces, conditions, figure source.
- Structures for 8 MOFs; 2 with stand-ins; 138 dataset CIFs for the unvalidated run.

Metrics

- Dominant face and habit against each record.
- Baselines: laboratory majority per MOF (paper held out); other MOFs (MOF held out).
- Wilson intervals; bootstrap over whole papers.

4. Implementation Details – Overview

Module-wise implementation and integration status

Module	Implementation	Status
Structure and symmetry	CIF reading, symmetry expansion, conventional cell	Integrated
Faces and shapes	Absences, d_{eff}, BFDH, families, deformed cells, any lattice	Integrated
Energies	Bond-count/UFF cleavage energies; CHGNet slabs (unvalidated)	Integrated
Benchmark, steering, access	Ten MOFs, 134 records; energy cut per face; open fraction per face	Integrated
Website and tests	27 pages; 15 jsdom suites (over 3 400 checks); Python tests	Functional

Environment and repository

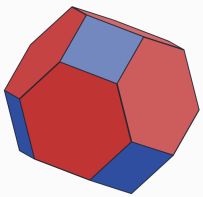
Python 3.11–3.13 (numpy, scipy, networkx, ase, spglib); JavaScript with three.js; CPU for every result shown; Colab GPU for CHGNet. `github.com/Radha-Krishna-19/fyp`

Benchmark (`run_benchmark_mofs.py`)

- **In:** CIF of each MOF; the laboratory CSV.
- **Out:** predicted face and habit; scores per model; MOF-level counts and paper bootstrap.
- Primitive cells are converted to the conventional cell first.

Steering and access

- Steering: closed forms for pure shapes; bisection on exact Wulff areas for the dominant face.
- Access: plane at the minimum-energy termination, 0.3 Å grid; checked against a 3D estimate.



HKUST-1 (published structure)

Predicted from the structure alone: **truncated octahedron**, mostly {111} faces (Wulff shape from unrelaxed bond-count surface energies with UFF van der Waals terms).

Laboratory records: **29**; dominant face reported: 7 × {100}, 13 × {111}; overall form: 20 compact, 1 plate, 2 rod.

Agreement: main face right in 13 of 20 records; compact / rod / plate right in 20 of 23.

Shape advisor page: HKUST-1 predicted shape and laboratory records.

6. Verification & Quantitative Results

Dominant face against laboratory records (eight MOFs with structures)

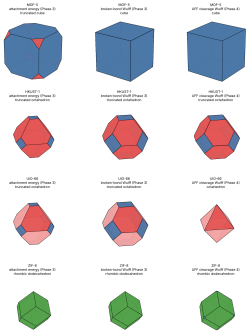
Model	No additive (32)	With additive (44)	All (76)
BFDH, structure only	18	21	39
Bond-count Wulff, structure only	**28**	17	45
Laboratory majority per MOF, paper held out	28	13	41
Majority of other MOFs, MOF held out	5	17	22

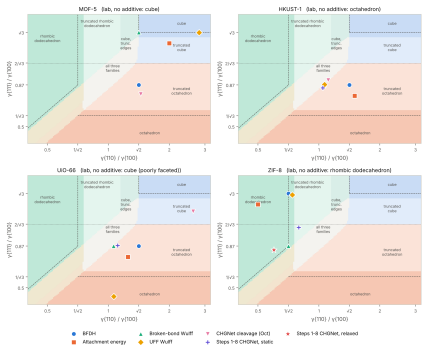
What this shows, and where it stops

- Without additives: right for **4 of 5 MOFs** (MOF-5 9/9, ZIF-8 11/12, HKUST-1 7/8, IRMOF-3 1/1; UiO-66 0/2); paper bootstrap 73–97%. As good as the laboratory majority, with no laboratory data.
- With additives no model is reliable. MOFs added after the model was fixed: 1 of 1 without, 1 of 7 with additives.
- Habit: 85 of 97 against 76 for “always compact”; Mg-MOF-74 rods, 9 of 10 crystals elongated.

7. Visual Results & Demonstration

Predicted shapes and the morphology map





Live demonstration

Website: upload a CIF on “From a CIF to a shape”; choose a MOF on “Shape advisor” to see its 3D shape, the energy cut needed for each face, and how open each face is to CO_2 and N_2.

8. Technical Justification, Design Trade-offs & Insights

Rationale, trade-offs and limits

Justification

- BFDH runs on any CIF; the bond-count energy adds what it lacks, with no fitting.
- No learned model: shapes exist for 12 MOFs; learning from other MOFs gives 5 of 32.

Edge cases

- Primitive cells converted; broken-symmetry files (IRMOF-8, P1 hMOF) flagged.
- Unclear additive labels listed; missing faces not scored.

Trade-offs

- Seconds per MOF, but unrelaxed energies without solvent or electrostatics.
- MOF-5's $\gamma_{111}/\gamma_{100} = 1.72$ is 0.7% below $\sqrt{3}$: its cube rests on one constant.

Limitations and future work

- DFT on six prepared slabs; relaxed CHGNet rerun.
- Gas access is geometric, not adsorption.
- Part I descriptor withdrawn: it follows graph size.

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Thank You

Questions & Viva Voce